IULIA RUSU

Ph.D. Student | Reinforcement Learning & Computational Neuroscience
Sharpee Lab | University of California, San Diego | Salk Institute for Biological Studies
University of California, San Diego **|** B.S. Physiology and Neuroscience | Ph.D. Biological Sciences (in progress)
Google Scholar: https://scholar.google.com/citations?user=G-KTa-QAAAAJ&hl=en

Work Experience

Ph.D. Student, Tatyana Sharpee Lab **September 2023-Present**

- Developing novel inverse reinforcement learning (IRL) methods for analysis of human behavior and decision-making
- Training and fine-tuning BERT-style encoders and latent space clustering analysis
- Implementing free-energy minimization RL methods to study information–reward tradeoffs in human and primate decision-making
- Analyzing neural data using dimensionality reduction methods inspired by Information Theory and Hyperbolic Geometry

Scientist I, Cell Therapy Engineering and Development, Genentech Inc. **September 2022-August 2023**

- Led team of NeoT process development scientists in isolating, activating, transfecting, and establishing CRISPR-engineered T cell product for patient tumor therapy
- Designed analytical protocols for QC analysis to ensure safety standards in final cell product are met
- Created and fostered relationships with Benchling to integrate pre-existing ELN data into new framework
- Conducted biomarker analysis in R on Multiple Sclerosis clinical data

Staff Research Associate, Kattah Lab, UC San Francisco **September 2019-August 2022**

- Conducted routine mouse colony management, tagging, tailing, weaning and genotyping of mice
- Opened Kattah Lab, set up lab space, ordered equipment, organized and maintained inventory
- Lab Management, up-keep of inventory of lab equipment and reagents
- Optimized protocols establishing, maintaining, and characterizing human and murine intestinal cell organoids
- Performed Western Blot, RNAscope, FACS, ELISA, Organoid culture, cloning and transfection to elucidate immunological mechanisms behind intestinal inflammation of mouse models of IBD
- Recruited healthy and Ulcerative Colitis (UC) patients for multi-omics study of UC
- Established 12 plex RNAscope protocol, and full supervised and unsupervised analysis pipeline for MIBI and RNAscope

Medical Neuroscience Research Assistant, Telese Lab, UC San Diego **September 2017-May 2019**

- Improved protocols for multiple immunohistochemistry and imaged the resulting tissue on a digital microscope for immunofluorescence imagery
- Analyzed and identified the genetic background of samples using epigenetic techniques in cell culture with immortal neuronal carcinoma cells
- Verified efficacy of antibodies using Western Blot, Blocking Peptide Western Blot, and shRNA transfection
- Presented in lab meetings and discussions about the future and goals of the lab team

Extracurricular Activities

Nucleate San Diego – Partnerships and Sponsorships Team Lead **September 2025- present**
Genentech Women in Science and Engineering (gWise) – Group Leader/Facilitator **September 2022-2023**
UCSD NCAA Women's Crew Team- Varsity **September 2017-June 2019**

- 2nd place in DII National Championships in 2018 and in a heavyweight 8 in WIRA in 2019

Awards

Pathways in Biological Sciences (PIBS) Fellowship **June 2024-June 2026**

Conferences & Presentations

American Physical Society- Denver, Colorado – Talk **March 2026**
Society for Neuroscience- San Diego, California – Poster **November 2025**
Society for Neuroscience-Chicago, Illinois- Poster **October 2024**
UCSF Immunology Retreat- Poster **June 2021**
University of California, San Diego Undergraduate Summer Research Program Symposium- Talk **August 2018**
Society for Neuroscience-San Diego, California- Poster **November 2018**
Society for Advancement of Chicanos/Hispanics and Native Americans in Science-Conference- Poster **October 2018**

Skills

Computational: PyTorch, Gymnasium
Experimental Techniques: qPCR, Western Blot, cell & organoid culture, RNA/DNA/protein extraction, cloning, ELISA, scRNA sequencing, Flow Cytometry, Immunofluorescence Microscopy, RNAscope,
Bio-Computational: HALO, QuPath, Cellxgene, Cellprofiler, Graphpad, FlowJo, Prism
Languages: Russian (native), French (A2)